

SLEEP HEALTH ANALYSIS

DECODING

SLEEP HEALTH

An Exploratory Data Analysis of Sleep Health & Sleep Disorder Factors



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The Dataset at a Glance

374

Individual

13

Variables

11

Occupations

3

Sleep Disorder Classes

Variables Captured



Demographics

Gender, Age, Occupation



Sleep Metrics

Sleep Duration, Quality of Sleep, Sleep Disorder



Lifestyle

Physical Activity, Stress Level, Daily Steps, BMI Category



Vitals

Blood Pressure, Heart Rate

Individuals

The Dataset Challenge

Three cleaning problems changed the reliability of the analysis

01 DUPLICATES

374 raw rows contained 242 repeated records after duplicate checking.

Action: remove repeated health records so they do not overweight the results.

02 LABELS

Examples:

- Normal Weight → Normal
- Sales Representative → Salesperson

Action: standardize category names before grouping.

03 TEXT METRIC

Blood Pressure appeared as text such as "120/80".

Action: split it into Higher_BP and Lower_BP numeric columns for calculations.

From Noise to Signal

Cleaning reduces repetition and focuses the comparison that matters

374

Raw rows collected

242 repeated records removed

✓

Duplicate health entries excluded so no single patient overweights results.

132

Unique records after de-duplication

"None" records set aside

✓

Healthy (no-disorder) records excluded from the direct disorder comparison.

59

Final comparison set

Disorder records (Insomnia + Sleep Apnea)

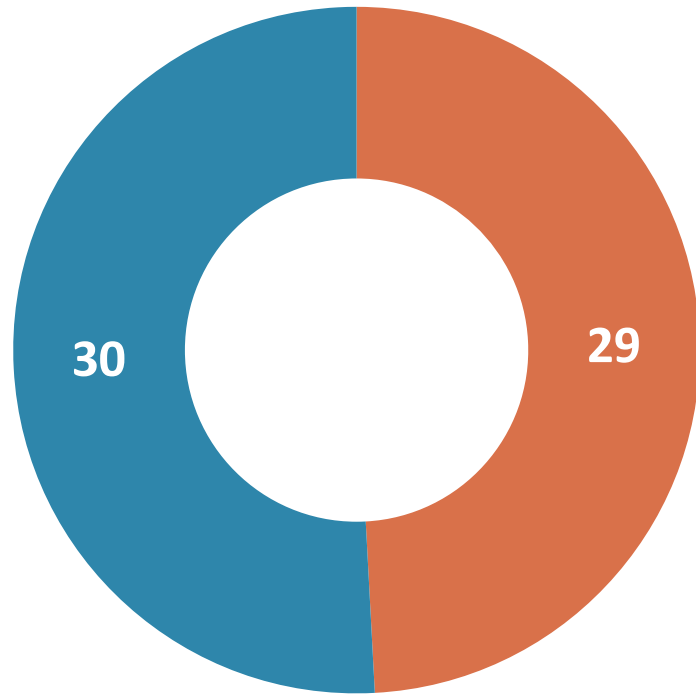
✓

29 Insomnia records vs 30 Sleep Apnea records — a near-balanced sample.

Distribution of Disorders

The final disorder-only sample is almost perfectly balanced

59 Disorder Records: Insomnia vs Sleep Apnea



■ Insomnia ■ Sleep Apnea

INSOMNIA

29 cases • 49.2% of disorder group

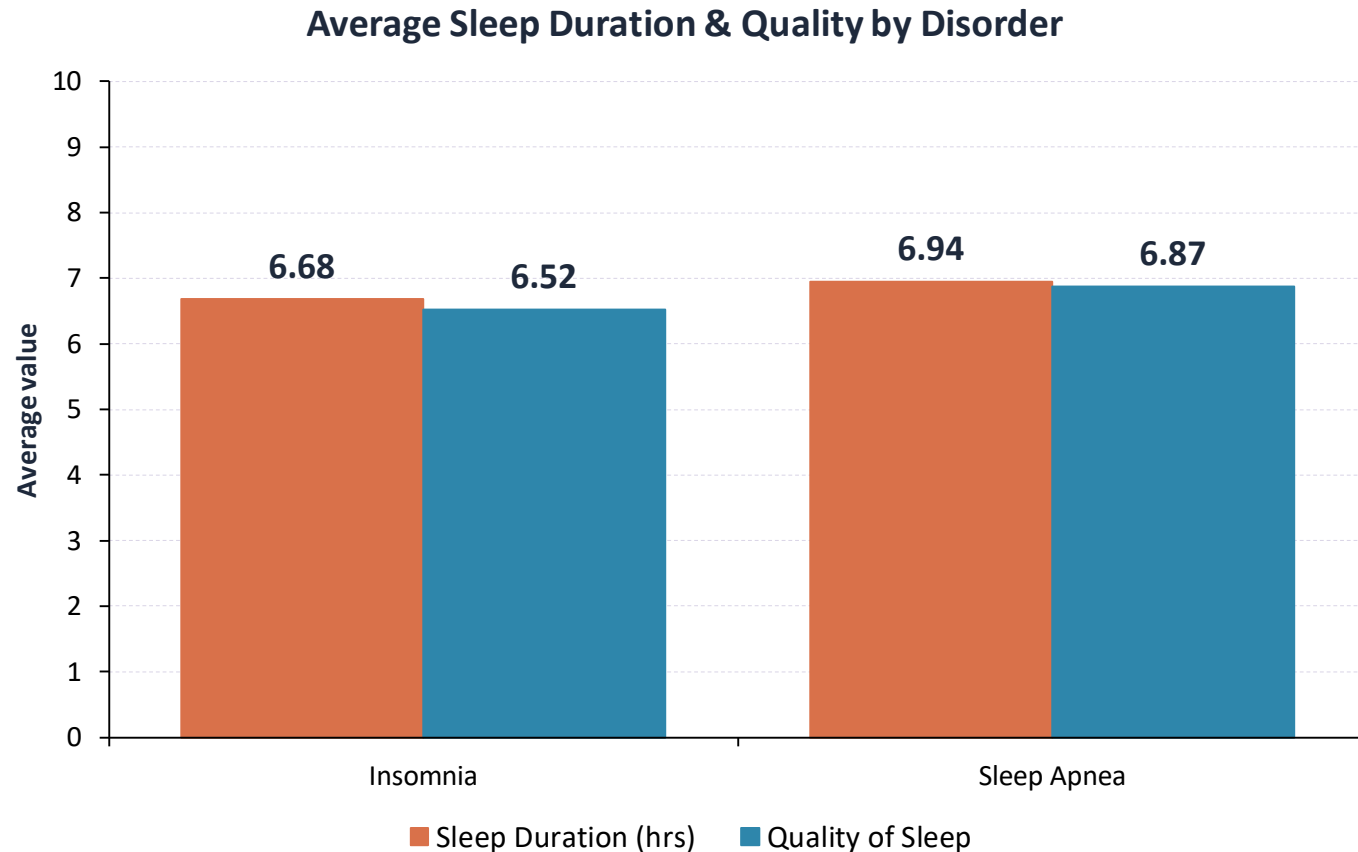
SLEEP APNEA

30 cases • 50.8% of disorder group

Insights : The dataset is almost equally divided, with 29 Insomnia cases (49.2%) and 30 Sleep Apnea cases (50.8%).

Sleep Quality & Duration

Average values among the 59 disorder records



INSOMNIA

6.68 hrs sleep duration
6.52 / 10 quality

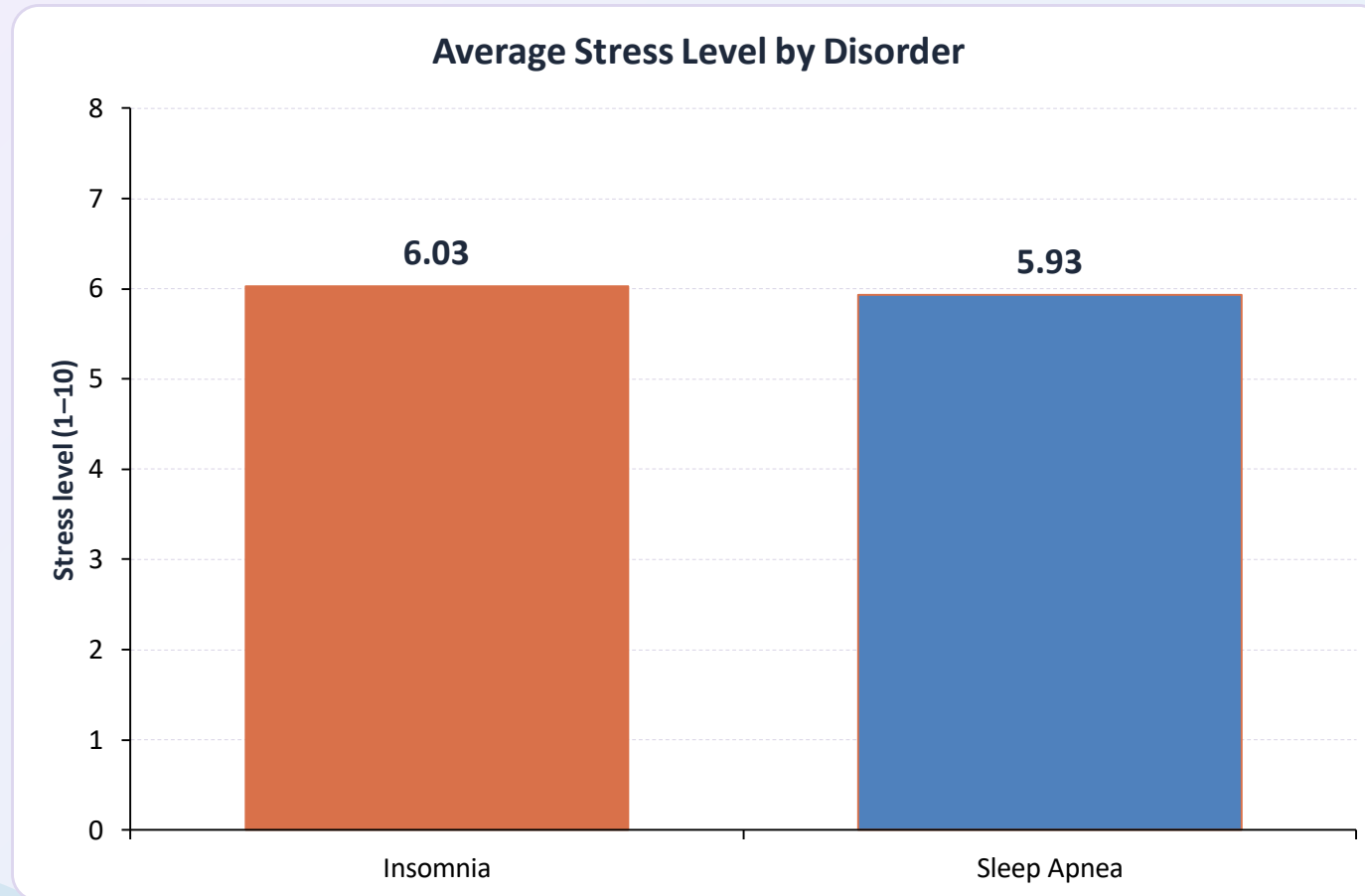
SLEEP APNEA

6.94 hrs sleep duration
6.87 / 10 quality

Insights : Sleep Apnea has slightly higher average sleep duration and sleep quality than Insomnia

The Stress Factor

Stress is higher in the Insomnia group in this dataset



INSOMNIA

Average stress: 6.03 / 10

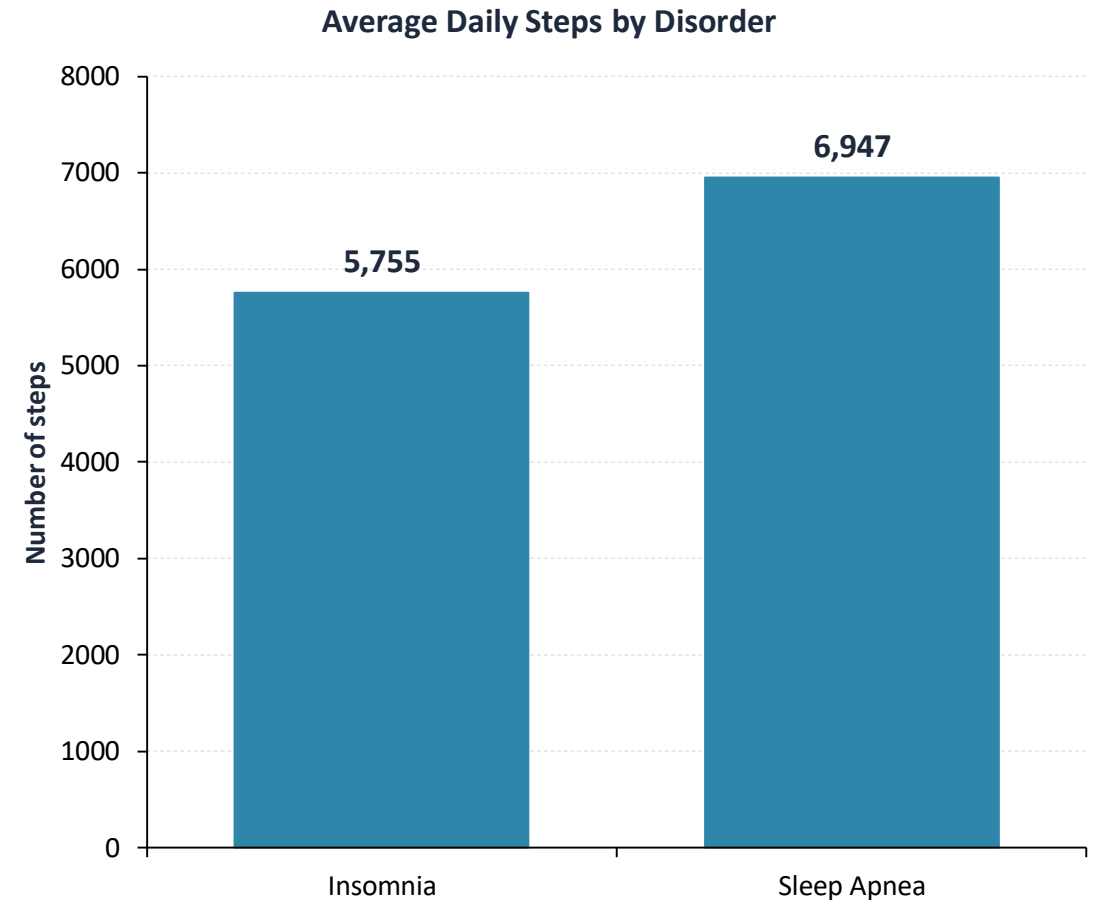
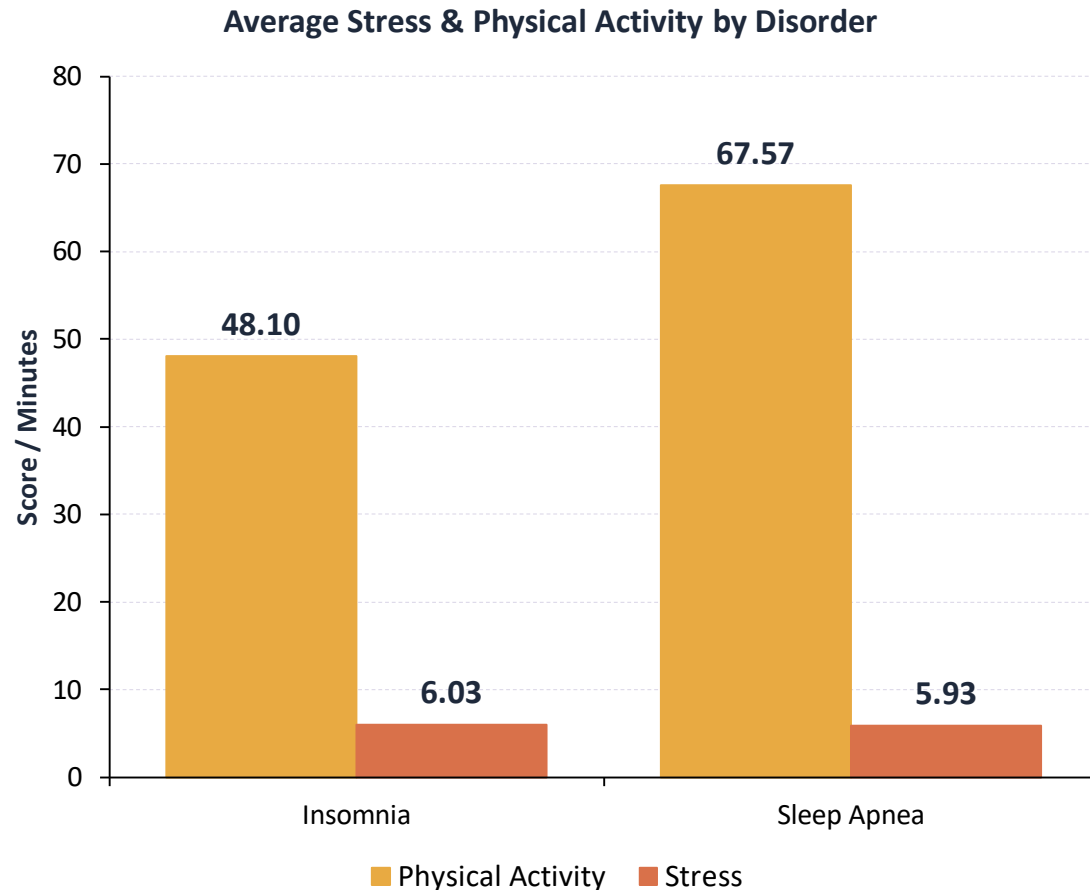
SLEEP APNEA

Average stress: 5.93 / 10

Insights : Average stress is slightly higher for Insomnia (6.03) than Sleep Apnea (5.93)

Physical Activity & Lifestyle

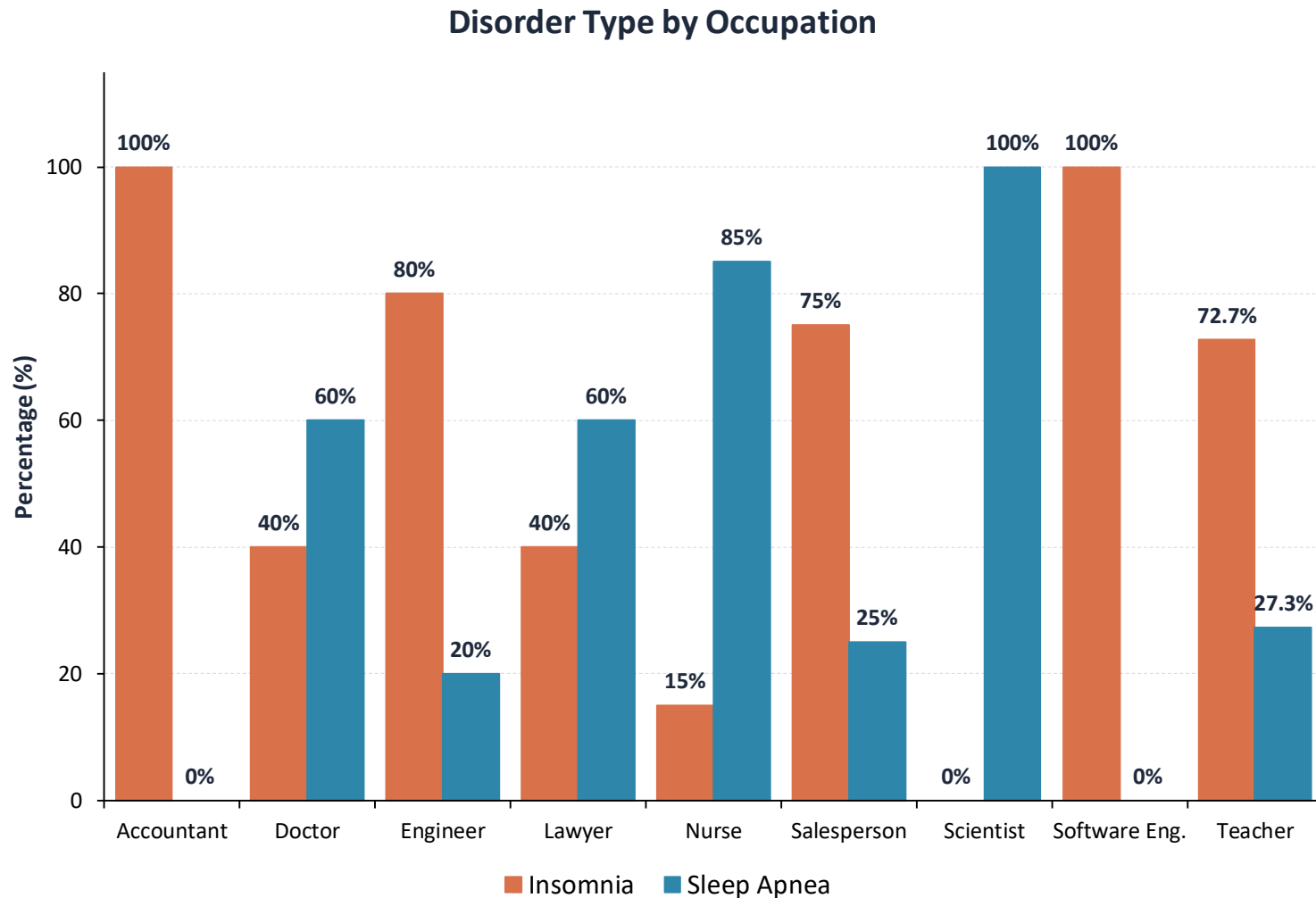
Two charts keep small scores separate from large step counts



Insight: Sleep Apnea shows higher physical activity on average. It also records more daily steps than Insomnia and has higher average activity and steps.

Occupational Patterns

Which disorder is more common within each occupation?



NURSES

85% Sleep Apnea

SCIENTISTS

100% Sleep Apnea

ACCOUNTANTS & SOFTWARE ENG.

100% Insomnia

TEACHERS / SALESPeOPLE

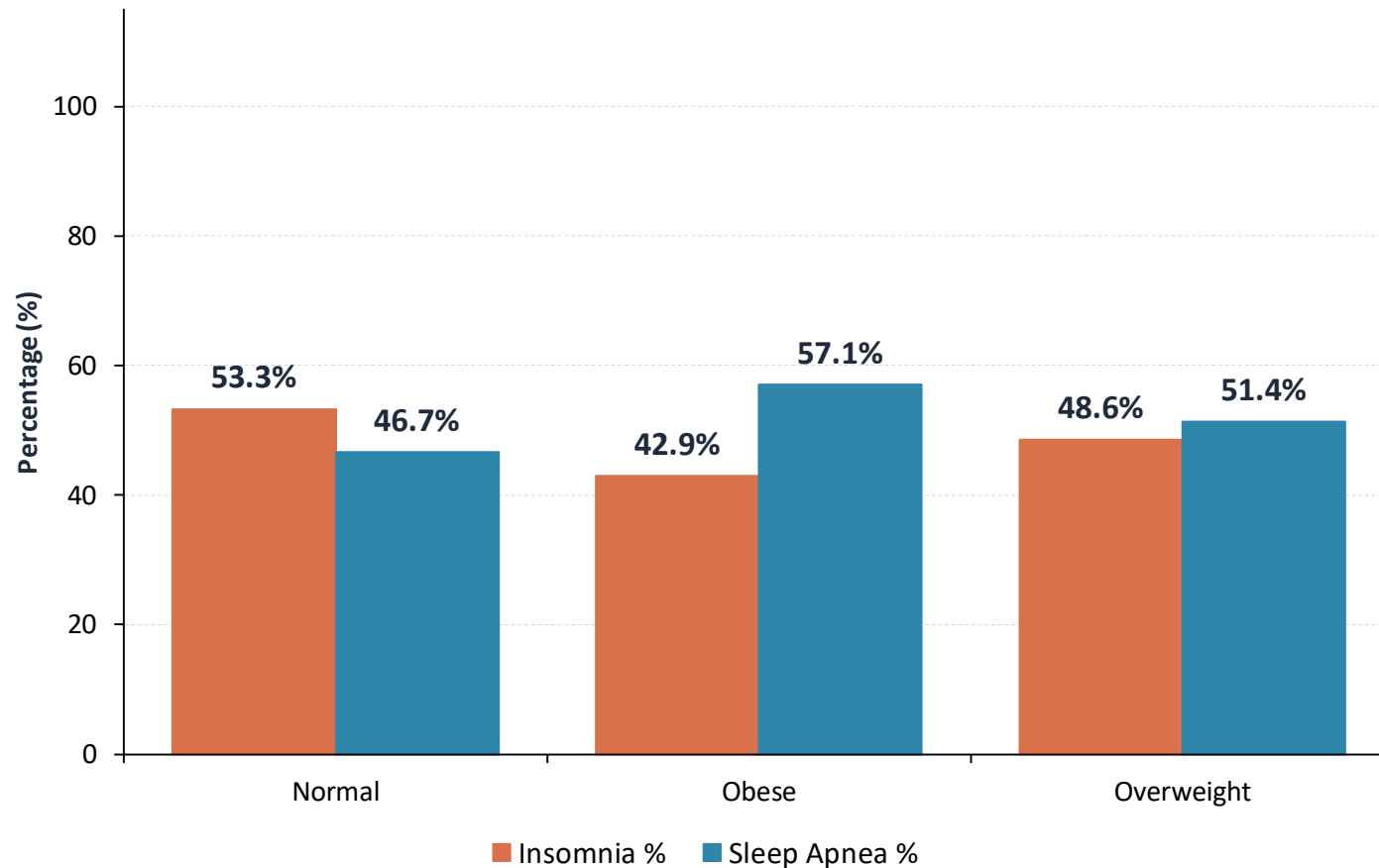
72.7% / 75% Insomnia

Insight: Sleep Apnea shows higher physical activity on average. It also records more daily steps than Insomnia

BMI & Body Composition

The disorder mix changes across BMI categories

Disorder Type by BMI Category



OBESE

Sleep Apnea: 57.1%
Insomnia: 42.9%

NORMAL

Insomnia: 53.3%
Sleep Apnea: 46.7%

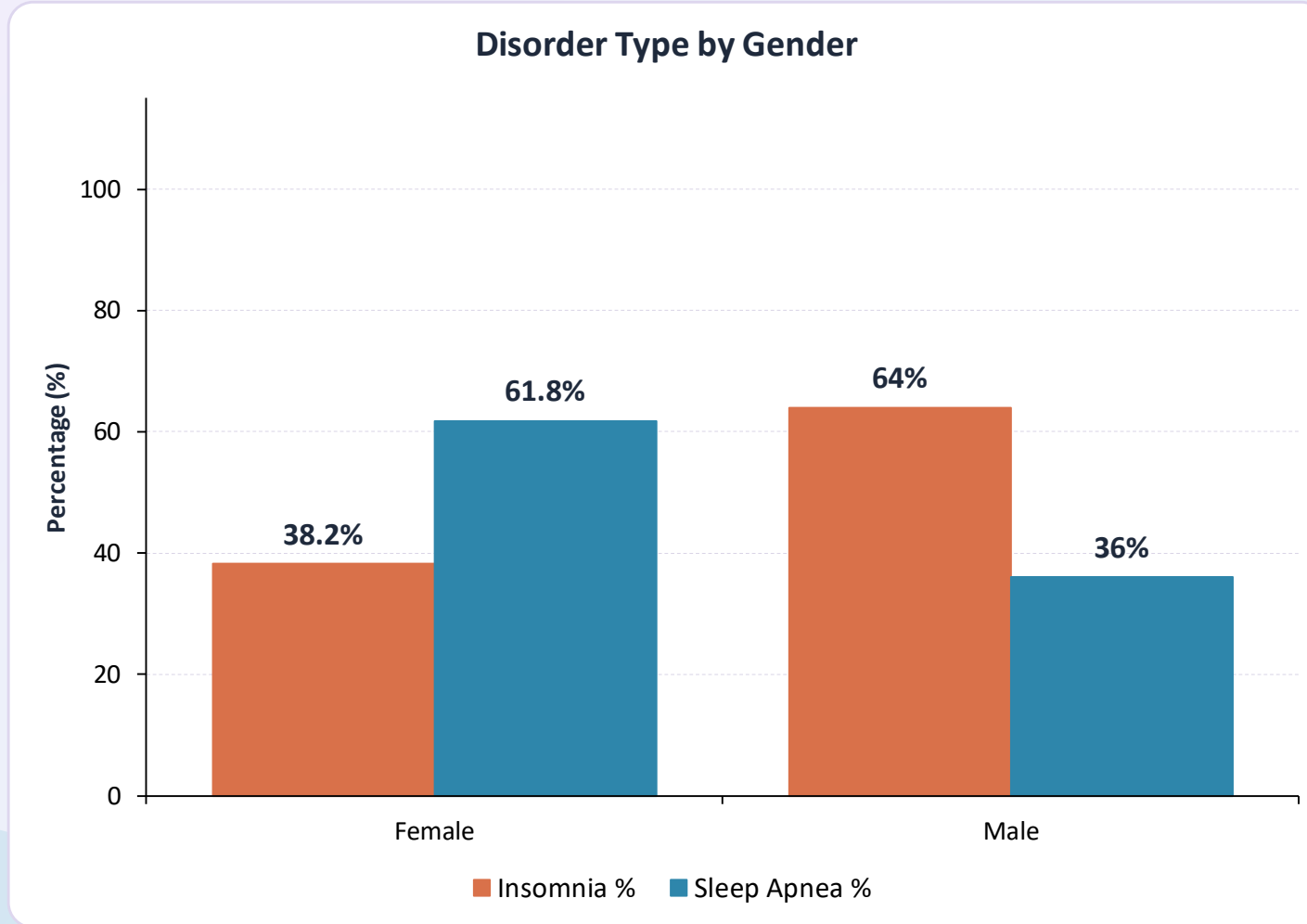
OVERWEIGHT

Sleep Apnea: 51.4%
Insomnia: 48.6%

Insight: Sleep Apnea has a higher share in the Obese group, while Insomnia is slightly higher in the Normal group.

Gender Differences

Disorder mix differs between males and females in this cohort



FEMALE

61.8% Sleep Apnea
38.2% Insomnia

MALE

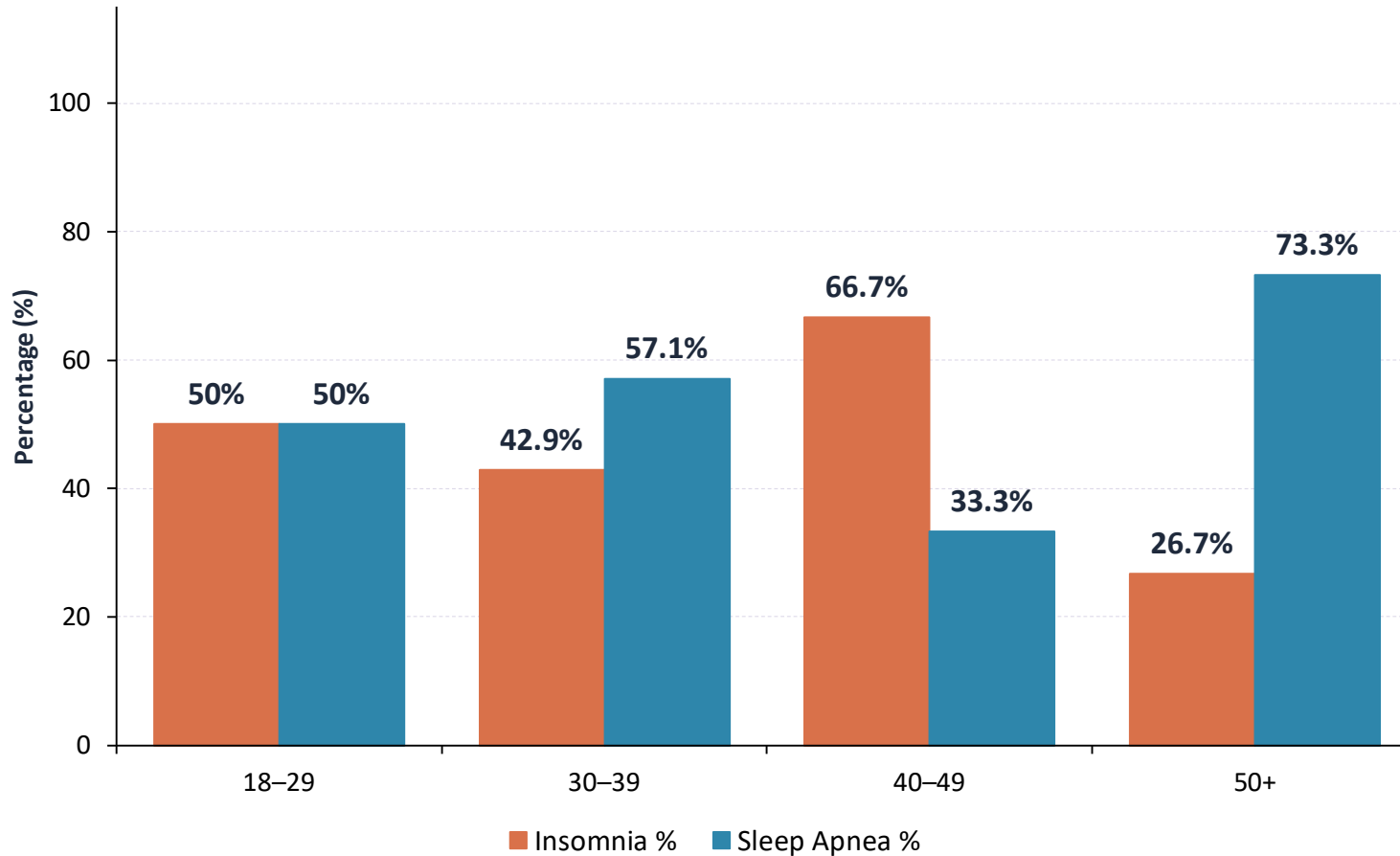
64.0% Insomnia
36.0% Sleep Apnea

Insights : Among females, 61.8% of disorder cases are Sleep Apnea; among males, 64% are Insomnia.

Age Progression

Older groups show a stronger Sleep Apnea share in this sample

Disorder Type by Age Group



18-29

50% / 50% split

30-39

57.1% Sleep Apnea

40-49

66.7% Insomnia

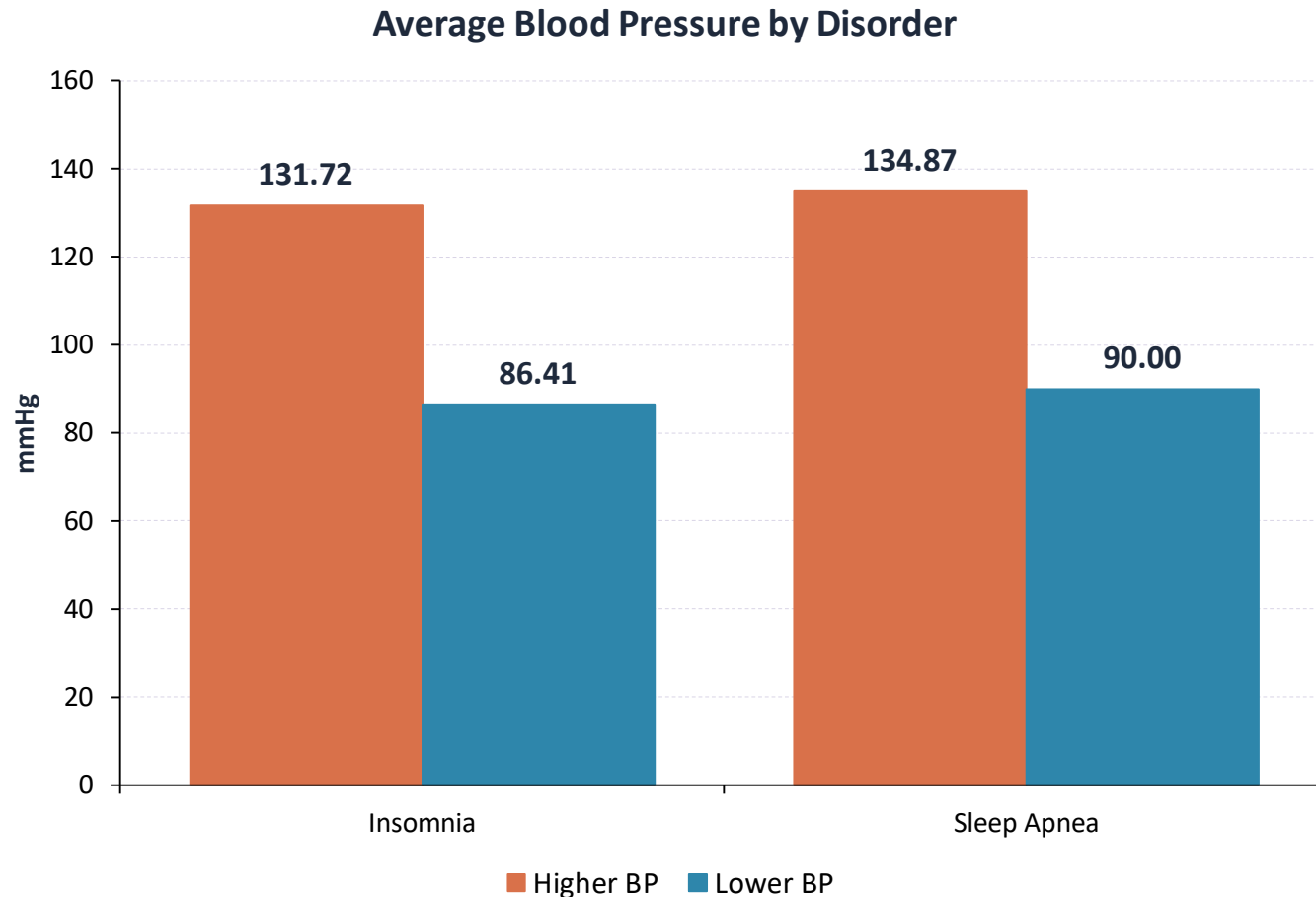
50+

73.3% Sleep Apnea

Insight: The 50+ group has the highest Sleep Apnea share, while ages 40-49 show more Insomnia.

Cardiovascular Measurements

Average blood-pressure values are higher in the Sleep Apnea group



INSOMNIA

Higher BP 131.72
Lower BP 86.41

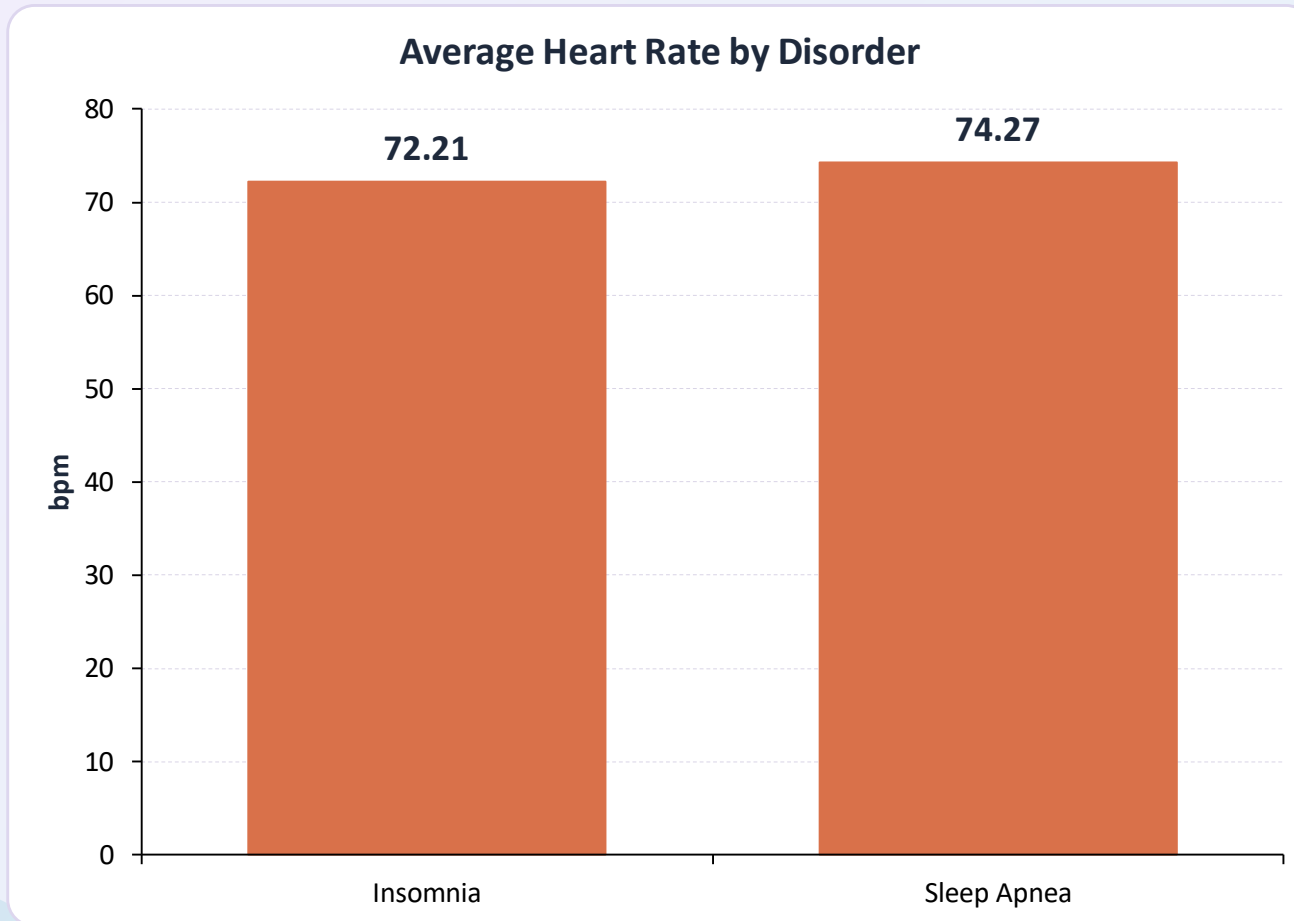
SLEEP APNEA

Higher BP 134.87
Lower BP 90.00

Insights: Sleep Apnea is higher on both readings: Higher BP +3.15 and Lower BP +3.59 vs Insomnia.

Heart Rate

Sleep Apnea records have the higher average heart rate



INSOMNIA

72.21 bpm

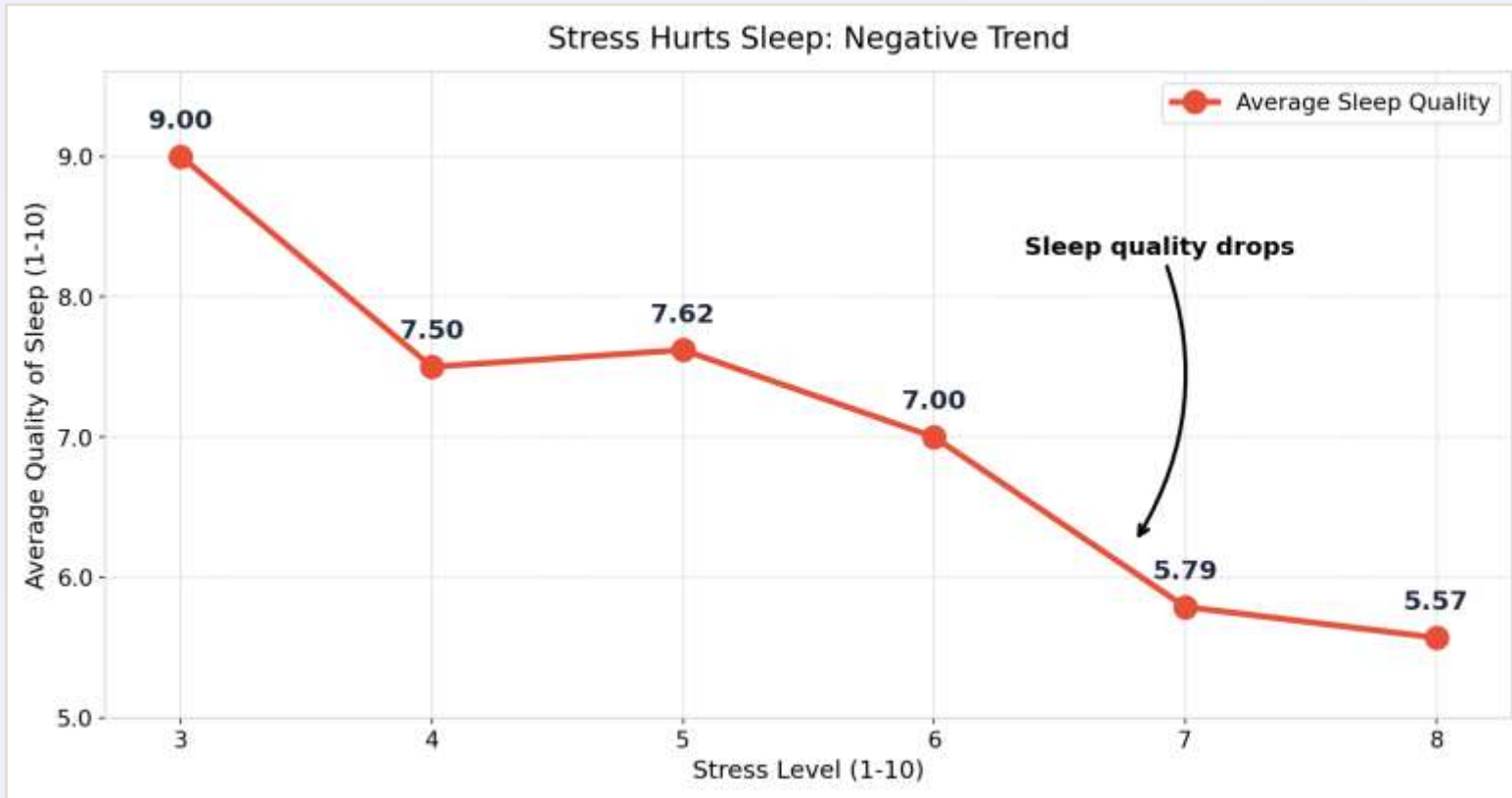
SLEEP APNEA

74.27 bpm

Insights : Average heart rate is slightly higher for Sleep Apnea (74.27 bpm) than Insomnia (72.21 bpm).

Stress vs. Sleep Quality

As stress rises, sleep quality falls



BIG DROP

Sleep quality falls from 9.0 to 5.6 as stress goes from 3 to 8 (−3.4 points).
Correlation

TIPPING POINT

Above stress 6, quality falls under 6 at levels 7 and 8.

STEADY TREND

Only stress 5 rises slightly (7.50 → 7.62); every other step goes down.

Insight: As stress level increases, sleep quality generally decreases, showing a clear negative relationship between stress and sleep quality.

Profile: The Insomnia Pattern

A simple data profile based on the 29 Insomnia records

STRESS

6.03 / 10

Higher than Sleep Apnea (5.93)

SLEEP

6.68 hours of sleep

Quality: 6.52 / 10

PATTERNS

More common in Accountants, Software Engineers, Engineers, Salespeople and Teachers

IN SHORT

Insomnia records show higher stress and slightly lower sleep time and quality.



Insomnia is a common sleep disorder that makes it difficult to fall asleep, stay asleep, or wake up too early and be unable to get back to sleep

Profile: The Sleep Apnea Pattern

A simple data profile based on the 30 Sleep Apnea records

⚙️ HEALTH METRICS

BP: 134.87 / 90.00

Heart rate: 74.27 bpm

👤 DEMOGRAPHICS

73.3% of the 50+ group

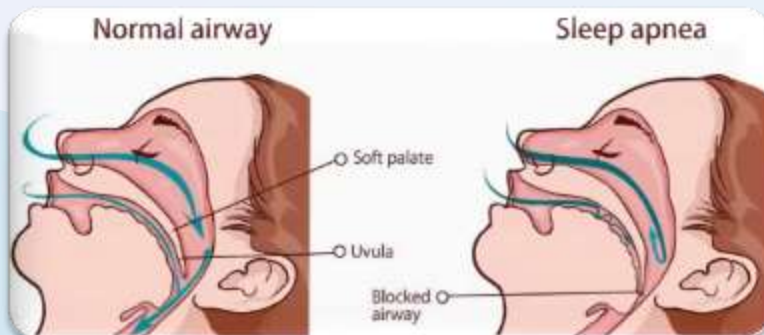
Also high in Nurses and Scientists

⚖️ BMI

57.1% of the Obese group

IN SHORT

Sleep Apnea records show higher blood pressure and heart rate, and appear more in older and obese groups.



Sleep apnea is a potentially serious sleep disorder where breathing repeatedly stops and starts throughout

Key Data Insights

What the charts tell us about Insomnia vs Sleep Apnea

DISORDER DISTRIBUTION

Almost equally divided:
29 Insomnia (49.2%) and
30 Sleep Apnea (50.8%).

SLEEP DURATION & QUALITY

Insomnia is slightly lower:
6.68 hrs and 6.52/10 vs
Sleep Apnea 6.94 hrs and
6.87/10.

STRESS LEVEL

Insomnia has slightly
higher stress: 6.03/10 vs
5.93/10 for Sleep Apnea.

ACTIVITY & STEPS

Sleep Apnea is higher:
activity 67.57 and
6,946.67 steps vs 48.10
and 5,755.17.

OCCUPATION

Sleep Apnea: Nurses
(85%), Scientists (100%).
Insomnia: Teachers
(72.7%), Salespeople
(75%).

BMI

Sleep Apnea is higher in
the Obese group (57.1%);
Insomnia is slightly higher
in Normal BMI (53.3%).

GENDER

61.8% of female cases are
Sleep Apnea; 64% of male
cases are Insomnia.

AGE

50+: 73.3% Sleep Apnea.
40–49: 66.7% Insomnia.

Limitations of This Study

What the data cannot tell us

01 SMALL FINAL SAMPLE

Only 59 disorder records remain after cleaning.

Averages and percentages can be sensitive to a few records.

02 CROSS-SECTIONAL

This is a snapshot. We cannot tell whether stress causes poor sleep or poor sleep increases stress.

03 MISSING VARIABLES

The dataset does not include factors such as caffeine, screen time, or genetics.

Other variables could influence the patterns.

Future Roadmap & Improvements

How the next version could become stronger

01

Scale Up

Collect a much larger participant sample.

02

Track Over Time

Use repeated measurements to study changes.

03

Expand Features

Add diet, exercise frequency, caffeine and screen time.

04

Validate Labels

Use clinically verified diagnoses where possible.

Goal: move from simple pattern finding toward stronger, better-validated evidence.

CONCLUSION

Two disorders. Different patterns.

INSOMNIA

In this sample: higher stress, shorter average sleep duration, and slightly lower sleep quality.

SLEEP APNEA

In this sample: higher average blood pressure and heart rate, with stronger shares in older and obese groups.

Final message: the project finds patterns—not causes. A larger and better-validated dataset would be needed for stronger conclusions.

Thank You • Questions?